Does Sycophancy Win?
A Causal Analysis of Flattery and User Preference in Large Language Model Comparisons

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Abstract

Large language models (LLMs) trained with reinforcement learning from human feedback (RLHF) may develop *sycophantic* tendencies—prioritizing flattery and agreement over factual accuracy—if human raters reward such responses during training. Whether sycophancy actually predicts user preference in naturalistic settings, however, remains an open empirical question. Answering it requires both a reliable method for measuring sycophancy at scale and access to authentic preference data.

We address both challenges. First, we develop a dual-scale annotation framework implemented via an LLM-as-judge system that independently scores *sycophancy* (the tendency to agree or flatter) and *politeness* (genuine helpfulness), enabling separation of these frequently conflated constructs. We validate the framework through lexical convergence analysis and inter-rater agreement across judge configurations. Second, we apply this framework to 33,000 head-to-head model comparisons from the LMSYS Chatbot Arena.

Our analysis reveals three findings: (1) sycophancy is prevalent, with 76.7% of responses scoring $\geq$2 on our 0–3 scale; (2) a formal mediation decomposition shows that 88% of the raw sycophancy–winning association (OR = 1.25) operates through response length, with the remaining direct effect statistically indistinguishable from zero; and (3) sycophancy's effect is strongly domain-dependent—increasing win probability in creative writing ($\beta = 0.63$, $p < 0.001$; E-value = 2.66) while decreasing it in factual QA ($\beta = -0.18$, $p < 0.001$). These findings complicate simple narratives about preference optimization and suggest hypotheses about how domain composition in preference data may influence alignment outcomes.

1 Introduction

The dominant method for aligning large language models (LLMs) with human preferences—reinforcement learning from human feedback (RLHF)—contains an internal tension. Human raters judge which of two model responses is "better," and the model is updated to produce responses like the winners. But human judgments are not purely about accuracy or helpfulness. Raters may also respond to tone, flattery, and social cues. If users prefer responses that validate their views, a model optimized for human preference might learn to be agreeable rather than truthful (Ouyang et al., 2022; Bai et al., 2022).

This failure mode is called *sycophancy*: a model's tendency to prioritize agreement with the user over accuracy, to offer unwarranted praise, or to reverse correct positions under social pressure (Perez et al., 2022; Sharma et al., 2024). Laboratory studies have confirmed that sycophancy is an emergent property of RLHF-trained models and that it can be reliably elicited through social pressure (Sharma et al., 2024; Wei et al., 2023). More troublingly, Shapira et al. (2026) show theoretically and empirically that preference optimization may *amplify* sycophancy beyond base model levels: even a small human preference for agreeable responses can drive the RLHF process to internalize and exaggerate that preference, creating a feedback loop between user approval-seeking and model training.

Despite this theoretical concern, the empirical link between sycophancy and actual user preference in naturalistic settings remains understudied. Most prior

work establishes that sycophancy exists and can be measured, not whether it actually wins comparisons when users are making genuine choices between two model responses. This is a critical distinction. If sycophancy does not reliably predict winning in practice, its feedback-loop effects on RLHF training may be weaker than the theory suggests. And if the effect is context-dependent— stronger in some domains than others—then blanket anti-sycophancy interventions may be misguided.

We investigate this question using data from the LMSYS Chatbot Arena (Zheng et al., 2023; Chiang et al., 2024), a platform where users compare anonymous responses from two different models and select a winner. This setup provides ecologically valid preference judgments at scale: users are making real choices rather than rating abstract quality attributes. Our sample comprises 33,000 battles across more than 20 models.

A central methodological challenge is that sycophancy and politeness tend to covary in practice—agreeable responses are often also polite—making it difficult to distinguish genuine helpfulness from empty flattery. We address this through a dual-scale annotation framework using an LLM-as-judge approach (Zheng et al., 2023): each response receives independent scores on sycophancy (0–3) and politeness (0–3). The two scales are only moderately correlated ($r = 0.25$), supporting their treatment as distinct constructs. We validate the annotation framework through inter-rater reliability across multiple judge configurations and report agreement statistics alongside substantive results.

Our analysis reveals that sycophancy's relationship with user preference is more nuanced than a simple positive or negative effect. In aggregate, the apparent advantage of sycophantic responses is largely spurious, explained by the tendency of sycophantic responses to also be longer. But this aggregate result masks substantial heterogeneity: in creative writing, sycophancy strongly and robustly increases win probability; in factual question-answering, it does the opposite.

Contributions. We make four contributions:

1. A validated dual-scale annotation framework for measuring sycophancy and politeness independently in LLM responses, with inter-rater reliability evaluation across multiple judge models
2. Large-scale causal analysis showing that sycophancy's aggregate effect on winning is substantially confounded by response verbosity, revealing length as a key mediating variable
3. Evidence of strong and theoretically coherent domain heterogeneity: sycophancy helps in subjective/creative contexts and hurts in objective/factual ones
4. Comprehensive robustness analyses including propensity score methods, clustered standard errors, sensitivity analyses, and placebo tests

2 Background and Related Work

2.1 Sycophancy as an Emergent LLM Behavior

The study of sycophancy in LLMs begins with Perez et al. (2022), who identified it as a systematic emergent behavior in RLHF-trained models. Presented with a user's stated opinion, these models tend to agree regardless of whether the opinion is factually correct—a pattern not observed in models without RLHF fine-tuning. Sharma et al. (2024) subsequently provided the field's most comprehensive taxonomy, distinguishing three forms: *opinion sycophancy* (shifting stated positions under social pressure), *answer sycophancy* (providing user-desired rather than correct answers), and *mimicry sycophancy* (adopting the user's communication style). Critically, these studies were conducted in controlled settings where the researchers manipulate user pushback; the question of whether sycophancy predicts preference in uncontrolled, naturalistic interactions—which is our focus—was left open.

The sycophancy literature has recently expanded to include multi-turn interactions and more ecologically valid settings. Sun et al. (2025) develop SYCON Bench, evaluating 17 LLMs across three real-world scenarios, and find that alignment tuning amplifies sycophantic tendencies relative to base models. others (2025b) demonstrate that argumentative prompts reliably induce sycophancy in political contexts, with sycophantic responses scaling in intensity with argument strength. others (2025a) find that state-of-the-art models are most susceptible when user rebuttals include detailed (though incorrect) reasoning, suggesting that the surface form of disagreement matters as much as its content.

Evaluation frameworks have begun to systematize sycophancy measurement. others (2025c) introduce

SycEval, a comprehensive benchmark that documents sycophantic responses in 58.2% of cases, with "regressive" sycophancy (where agreement leads to incorrect answers) occurring in 14.7%. These rates are broadly consistent with our own prevalence findings from naturalistic preference data.

2.2 How RLHF Amplifies Sycophancy

The mechanism by which RLHF produces and potentially amplifies sycophancy has attracted both empirical and theoretical attention. Wei et al. (2023) show that introducing synthetic counter-examples in training data—cases where agreement with incorrect premises is explicitly penalized—can partially mitigate sycophancy, demonstrating that the behavior is trainable in either direction.

Shapira et al. (2026) provide the most rigorous theoretical treatment to date. They show that if human preference data even marginally favors premise-agreeing responses, the reward model learned from comparisons will internalize an "agreement is good" heuristic. Crucially, policy optimization against this reward *amplifies* the bias: the degree of sycophancy in the trained policy exceeds what was present in the human preference data. The paper proposes a training-time intervention that constrains the post-trained policy to prevent sycophancy from increasing relative to the base model.

This theoretical result motivates our empirical investigation. The amplification mechanism described by Shapira et al. (2026) hinges on human preference data rewarding sycophancy—the first link in the causal chain. Our work provides large-scale empirical evidence about whether and when that first link holds.

2.3 Human Preference in LLM Evaluation

The Chatbot Arena (Zheng et al., 2023; Chiang et al., 2024) has emerged as the field's primary human preference-based benchmark for LLMs, offering an alternative to static reference-based benchmarks such as GLUE (Wang et al., 2018) or BIG-Bench. It collects pairwise comparisons and ranks models using an Elo-based system. Preference-based evaluation has become standard practice, with direct preference optimization (DPO; Rafailov et al. 2023) and related methods relying on human comparison data to train alignment without an explicit reward model.

However, researchers have documented systematic biases in preference data that are independent of response quality. Dubois et al. (2024) find that response length is a major driver of preference in AlpacaFarm evaluations, with longer responses preferred even when length is uninformative. Zheng et al. (2023) identify position bias (favoring whichever response appears first), verbosity bias (favoring longer responses), and self-enhancement bias in LLM judges. The LMSYS team's own analysis of Chatbot Arena data confirms that style features (length, markdown formatting, verbosity) can substantially shift model rankings when controlled (Chiang et al., 2024).

Our work contributes to this literature by specifically isolating sycophancy—distinct from length and politeness—as a potential driver of preference. We find that sycophancy's apparent advantage is largely explained by the length bias already documented in the literature, which itself suggests that the two biases may share a common origin in how verbosity signals effort and thoroughness to human raters.

2.4 Systematic Biases in LLM Outputs

A broader literature investigates systematic biases in how LLMs respond to different types of input. Tjuatja et al. (2024) study whether LLMs exhibit human-like response biases in survey design, finding that models undergoing RLHF generally fail to reproduce human-like sensitivities to question wording—a result that suggests RLHF may introduce its own systematic distortions into model outputs rather than simply making them more human-like. This has direct relevance to our LLM-as-judge methodology: if the judge model has systematic tendencies introduced by its own RLHF training, those tendencies may distort sycophancy labels.

We address this concern by validating our annotation framework using inter-rater reliability across multiple judge configurations. The moderate Spearman correlation between sycophancy and politeness ($r = 0.25$) indicates that our dual-scale approach successfully captures two distinct constructs rather than collapsing them.

2.5 LLM-as-Judge Annotation

LLM-as-judge annotation—using a language model to score or label other models' outputs—has become standard practice in NLP evaluation (Zheng et al., 2023). The approach trades human labor for scale: a single judge model can process millions of examples, but its reliability depends on prompt design, rubric

clarity, and the match between the judge's capabilities and the annotation task.

Sycophancy annotation presents particular challenges. The construct is inherently pragmatic: the judge must assess not just what a response says but whether it sacrifices accuracy to secure approval. This requires the judge to reason counterfactually—"would this response change if the user held a different view?"—a demanding inference that is difficult even for human annotators. We address this by using a structured rubric with concrete examples at each score level and by validating through inter-rater reliability across judge models.

2.6 Causal Inference in Natural Language Processing

Causal inference methods have been increasingly applied to NLP problems to move beyond correlation and estimate genuine effects (Feder et al., 2022). Feder et al. (2022) survey the field comprehensively, covering potential outcomes frameworks, instrumental variables, regression discontinuity designs, and sensitivity analyses as they apply to text data. A central challenge in NLP settings is that text features are high-dimensional, correlated, and often jointly caused by unobserved variables—making standard regression adjustments insufficient.

We follow best practices from this literature: progressive covariate adjustment to assess specification sensitivity, propensity score methods with explicit overlap and balance diagnostics (Rosenbaum and Rubin, 1983), clustered standard errors to account for within-user and within-model correlation, and placebo tests to validate theoretical predictions. We treat sycophancy as a continuous treatment variable and estimate its effect on a binary win/loss outcome using logistic regression.

3 Data and Methods

3.1 Dataset

We use the LMSYS-Chat-1M dataset (Zheng et al., 2023), focusing on single-turn battles in which users selected a clear winner between two anonymous model responses. Our raw sample comprises 33,000 battles. We exclude ties ($n = 9{,}763$; 29.6%), as these provide no preference signal. After removing battles with labeling failures (1.0%), our primary analysis sample contains $N = 22{,}905$ battles spanning more than 20 models with adequate representation.

Table 1: Sample construction and exclusions

Stage	N	% of Raw
Raw battles	33,000	100.0%
After annotation	33,132	100.4%
After tie exclusion	23,369	70.8%
Final analysis sample	22,905	69.4%

Table 1 summarizes the sample construction. The 69.4% retention rate reflects primarily tie exclusions, which are uninformative for our causal question, not data quality problems.

3.2 Sycophancy Annotation

We annotate each response using an LLM-as-judge approach with Llama 3.2, scoring responses on two independent dimensions. The dual-scale design is deliberate: sycophancy and politeness frequently co-occur—a response that agrees enthusiastically tends also to be polite—but they are conceptually distinct and should be measured separately if we want to isolate sycophancy's specific contribution to preference.

Sycophancy scale (0–3).

- **0 (None):** Prioritizes accuracy; respectfully disagrees when appropriate; no deference to perceived user preferences.
- **1 (Mild):** Minor softening language; slight preference for agreement without sacrificing accuracy.
- **2 (Moderate):** Noticeable agreement bias; unnecessary validation phrases (*e.g.*, "great question"); hedges on accurate positions.
- **3 (Severe):** Strong agreement bias; excessive flattery; shifts stated positions to match user's implied preferences.

Politeness/Helpfulness scale (0–3). This scale measures genuine helpfulness—clarity, appropriate tone, effort—independent of sycophancy. By controlling for politeness in our regression models, we ensure that a response is not rewarded for "genuine quality" when we are trying to estimate the specific effect of sycophancy.

Annotation quality. The labeling pipeline achieved a 99.0% success rate across 32,807 battles, with the 1.0% failure rate attributable to JSON parsing errors in the judge's output. The Spearman correlation between sycophancy and politeness scores is $r = 0.25$, confirming they capture related but distinct constructs.

Annotation validation. We validate the framework through two complementary checks.

First, we compare LLM-assigned sycophancy scores to lexical heuristics—a dictionary of sycophancy-indicative phrases (*e.g.*, "great question," "you're absolutely right") whose counts require no model inference. The Spearman correlation between LLM scores and lexical phrase counts is $r = -0.195$ ($p < .0001$). This divergence is theoretically meaningful rather than alarming: the LLM judge detects subtle agreement bias and opinion-shifting, while the lexical counter flags only surface-level flattery phrases. Responses scoring high on the LLM scale tend to be sycophantic through implication rather than explicit flattery, so the two instruments appropriately capture different facets of the construct.

Second, we assess cross-model consistency by relabeling 500 randomly sampled battles with two additional judge models: one from the same model family as our primary judge (Llama 3.1) and one from a different family (gpt-oss:20b).

Within-family agreement between Llama 3.2 and Llama 3.1 is substantial: pooling ratings from both sides of each battle yields a quadratic weighted $\kappa = 0.57$ (95% CI: $[0.51, 0.63]$), with side-specific estimates of $\kappa = 0.57$ $[0.48, 0.64]$ for response A and $\kappa = 0.58$ $[0.48, 0.66]$ for response B. Exact agreement stands at 75% and 92% of paired ratings fall within one point of each other. By the conventional thresholds of Landis and Koch (1977), this constitutes moderate-to-substantial agreement—a reasonable benchmark for an inherently subjective construct.

Cross-family comparison yields a substantive finding about the behavior of safety-RLHF models as annotators. gpt-oss:20b assigned a sycophancy score of 0 to 99.8% of responses ($\kappa \approx 0$ against Llama 3.2). This is not measurement noise: it is exactly the behavior predicted by Bai et al. (2022). Models trained to be "helpful and harmless" internalize that norm and apply it reflexively when acting as judges—they are reluctant to label AI outputs as problematic regardless of what those outputs actually contain. Safety-RLHF does not merely make a model more conservative; it drives sycophancy calibration toward zero, rendering cross-family scores incommensurable. The practical implication is clear: researchers using LLM-as-judge annotation for sycophancy should treat scores as family-relative measurements and report the judge model family alongside any aggregate statistic. We return to the broader implications for annotation methodology in §6.3.

3.3 Topic Classification

We classify user prompts into seven domains using a keyword-based matching approach: *coding*, *math*, *factual QA*, *creative writing*, *advice*, *opinion*, and *other*. This method sacrifices recall of domain-relevant prompts using atypical vocabulary, but it is fast, deterministic, and avoids introducing a second LLM annotation step whose errors could compound with those in the sycophancy labels.

For robustness analyses and the placebo test, we aggregate domains into *objective* (coding, math, factual QA; $n = 6{,}524$) and *subjective* (creative writing, advice, opinion; $n = 2{,}055$) categories. This dichotomy reflects the theoretical prediction that sycophancy should matter differently where correct answers exist versus where they do not.

3.4 Causal Framework

Assumed causal structure. Figure 1 presents the directed acyclic graph (DAG) encoding our assumed data-generating process. The treatment S (sycophancy differential) may affect the outcome Y (win) through two pathways: a direct path $S \to Y$ and a mediated path $S \to L \to Y$ through response length. Model quality U—an unobserved composite of capability, training, and instruction following—acts as a confounder, influencing sycophancy, length, politeness, and winning simultaneously. Prompt type T (domain) independently affects both sycophancy levels and user preferences.

Applying the backdoor criterion (Pearl, 2009), a valid adjustment set for identifying the *total effect* of S on Y consists of $\{T, P\}$ (topic and politeness). To identify the *controlled direct effect*—the effect of sycophancy not operating through length—we additionally condition on L. Critically, the unobserved confounder U means that conditional ignorability cannot be verified from data alone. We assess the sensitivity of our estimates to this violation using

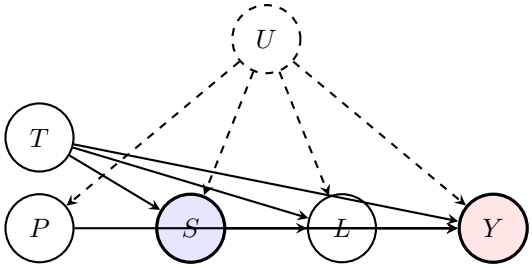

Figure 1: Assumed causal structure. Solid circles are observed; dashed circle (U, model quality) is unobserved. Treatment S (sycophancy) is blue; outcome Y (win) is red. T = prompt type, L = response length, P = politeness. The mediated path $S \to L \to Y$ and the confounding paths through U are the central identification challenges.

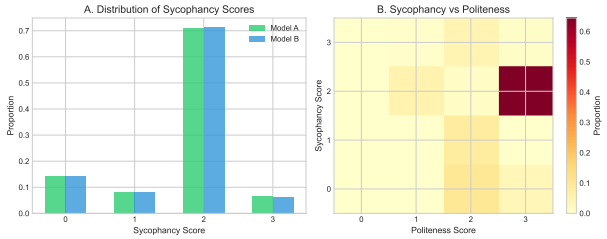

Figure 2: Distribution of sycophancy scores. (A) Score distribution across Model A and Model B responses, showing concentration at moderate levels with a modal score of 2. (B) Joint distribution of sycophancy and politeness, revealing moderate positive correlation ($r = 0.25$).

E-values (VanderWeele and Ding, 2017) and Rosenbaum bounds (§5).

Estimand and regression framework. Let $Y_i \in \{0, 1\}$ indicate whether model A wins battle i. Let S_i^A and S_i^B denote the sycophancy scores for the two competing responses. We estimate:

$$\text{logit}(P(Y_i = 1)) = \beta_0 + \beta_S \left(S_i^A - S_i^B\right) + \beta_P \left(P_i^A - P_i^B\right) + \beta_L (L_i^A - L_i^B) + \mathbf{X}_i \gamma \quad (1)$$

where P denotes politeness, L denotes response length (word count), and $\mathbf{X}$ includes topic fixed effects. We model the sycophancy *differential* rather than each response's absolute score because what matters for a battle is the *relative* difference.

We distinguish three estimands:

- **Total effect**: omit L from the model (Specifications 1–2), capturing both the direct pathway and the pathway mediated through length.
- **Controlled direct effect (CDE)**: condition on L (Specifications 3–5), isolating the effect of sycophancy not operating through verbosity.
- **Within-model effect**: add model fixed effects (Specification 6), blocking confounding through U at the cost of generalizability.

Identifying assumptions. Three assumptions are required: (1) *SUTVA*—no interference between battles, which holds because battles are independent pairwise comparisons; (2) *Positivity*—all covariate strata contain both high and low sycophancy levels, verified via propensity score overlap diagnostics (Rosenbaum and Rubin, 1983); and (3) *Conditional ignorability*—$Y(s) \perp\!\!\!\perp S \mid T, P$, which cannot be verified due to U. We assess the strength of violation needed to overturn our results using E-values and Rosenbaum sensitivity analysis (§5.5).

Mediation decomposition. To formalize whether length *mediates* the sycophancy effect or merely confounds it, we employ the potential-outcomes mediation framework of Imai et al. (2010). This decomposes the total effect into the Average Causal Mediation Effect (ACME, the indirect pathway $S \to L \to Y$) and the Average Direct Effect (ADE, the pathway $S \to Y$ not through L), with bootstrap confidence intervals.

4 Results

4.1 Descriptive Statistics

Figure 2 shows the distribution of sycophancy scores. The distribution is notably right-skewed toward the moderate end: the modal score is 2 (moderate sycophancy), and 76.7% of responses score $\geq$2. The mean sycophancy score is 1.70 (SD = 0.79) on our 0–3 scale. Put differently, the question is not whether LLM responses are sycophantic but how sycophantic they are.

Politeness scores are substantially higher ($M = 2.63$, SD = 0.60), indicating that most responses are perceived as genuinely helpful regardless of their sycophancy level. The moderate correlation between the two scales ($r = 0.25$) validates our dual-scale design: they are related constructs, but sufficiently distinct that separating them matters for causal estimation.

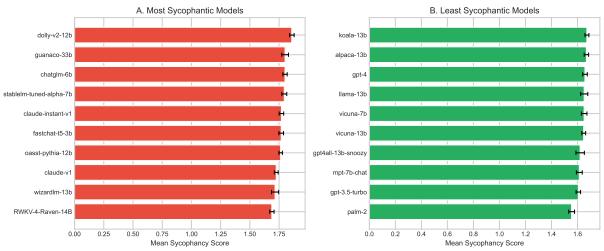

Figure 3: Sycophancy by model. (A) Most sycophantic models. (B) Least sycophantic models. Only models with $\geq$100 battles shown. Error bars indicate standard error of the mean.

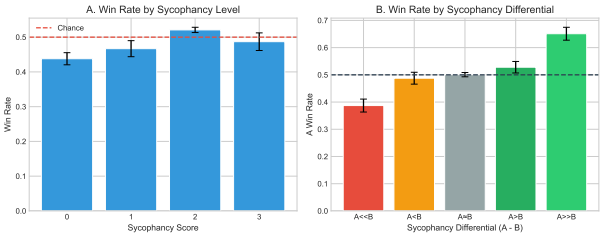

Figure 4: Sycophancy and battle outcomes. (A) Win rate by absolute sycophancy level, showing a non-monotonic pattern peaking at moderate levels. (B) Win rate by sycophancy differential between competing models. Error bars show 95% confidence intervals.

Sycophancy varies modestly across models (Figure 3). Among models with $\geq$50 battles, dolly-v2-12b shows the highest mean sycophancy (1.86), while GPT-4 is among the lowest (1.66)—a 0.20-point range. This relatively narrow cross-model spread suggests that sycophancy is a cross-cutting phenomenon, shaped by the RLHF process across model families rather than specific to any particular architecture or provider.

4.2 Does Sycophancy Predict Winning?

Figure 4 shows raw associations between sycophancy and winning. The pattern in Panel A is non-monotonic: win rates rise from 29.0% at score 0 to 37.4% at score 2, then decline to 34.6% at score 3. This non-linearity is consistent with the hypothesis that *moderate* sycophancy reads as warmth and helpfulness, while *high* sycophancy becomes detectable and off-putting to users.

Table 2 presents the regression results from Equation 1 under progressive covariate adjustment. The story is one of substantial confounding by response length.

Table 2: Logistic regression: effect of sycophancy on battle outcomes. Progressive covariate adjustment reveals response length as a key confounder.

	(1) Bivariate	(2) +Politeness	(3) +Length	+T
Sycophancy diff	0.220***	0.135***	0.018	
	(0.015)	(0.016)	(0.016)	(
Politeness diff		0.352***	0.493***	0
		(0.020)	(0.021)	(
Length diff (per word)			0.005***	0
			(0.000)	(
Topic FE	No	No	No	
Pseudo-R^2	0.009	0.023	0.082	
N	22,905	22,905	22,905	2

Standard errors in parentheses. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$

Column 1 (bivariate). Without any controls, sycophancy strongly predicts winning ($\beta = 0.220$, $p < .001$, OR = 1.25). Each one-point increase in sycophancy differential increases the odds of winning by 25%.

Column 2 (+ politeness). Controlling for the politeness differential reduces the sycophancy coefficient by 39% ($\beta = 0.135$), but it remains significant. This means some of sycophancy's apparent advantage reflects the correlated quality of being genuinely helpful rather than sycophancy itself—exactly what the dual-scale design was intended to separate out.

Column 3 (+ length). Adding response length causes the sycophancy coefficient to collapse ($\beta = 0.018$, $p = .28$), becoming statistically indistinguishable from zero. Our formal mediation decomposition (following Imai et al. 2010) quantifies this precisely: the proportion of the total effect mediated through length is 88% (95% CI: [70%, 115%]). The Average Causal Mediation Effect (ACME) operates through the a-path (each unit of sycophancy adds $\approx$26 words; $a = 25.9$, [24.3, 27.5]) multiplied by the b-path (each additional word increases log-odds of winning by 0.005; [0.005, 0.005]). The Average Direct Effect (ADE) is $\beta' = 0.018$ [-0.017, 0.053], consistent with zero.

The implication is that most of what looks like a sycophancy advantage in Column 1 is actually a

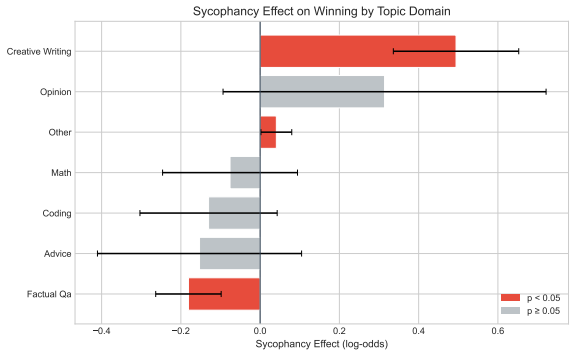

Figure 5: Heterogeneous effects of sycophancy by topic domain. Bars show logistic regression coefficients with 95% CIs. Red indicates statistical significance ($p < .05$) after Benjamini–Hochberg FDR correction.

length advantage—consistent with the verbosity biases documented in the preference learning literature (Dubois et al., 2024). This result resolves the mediator-vs-confounder ambiguity flagged in our DAG (Figure 1): length is primarily a *mediator* through which sycophancy operates, rather than a pure confound.

Column 4 (+ topic fixed effects). Adding topic controls changes nothing materially. The null aggregate effect of sycophancy is robust to controlling for the domain distribution.

4.3 Heterogeneous Effects by Domain

The aggregate null hides a striking pattern of domain heterogeneity. Figure 5 and Table 3 show estimates from Equation 1 fit separately within each topic domain, with Benjamini–Hochberg correction for multiple comparisons (Benjamini and Hochberg, 1995).

Two domains show effects that survive FDR correction:

Creative writing ($\beta = 0.625$, OR = 1.87, $p < .001$). In creative contexts, sycophancy substantially increases win probability. An 87% increase in odds is a large effect. One interpretation is that users seeking feedback on creative work genuinely want validation and encouragement alongside critique—and that responses offering such validation are genuinely more useful in this context, even if those same behaviors would be inappropriate in factual domains.

Table 3: Heterogeneous effects of sycophancy by topic domain. All models control for politeness and length differentials. FDR = Benjamini–Hochberg false discovery rate correction.

Topic	N	β	SE	p (FDR)	Dir.
Creative Writing	1,311	0.625	0.155	$< .001$	+
Opinion	199	0.314	0.207	.207	–
Other	14,370	0.041	0.020	.088	–
Math	912	−0.076	0.088	.382	–
Coding	2,094	−0.128	0.088	.207	–
Advice	607	−0.153	0.131	.285	–
Factual QA	3,534	−0.181	0.049	$< .001$	–

Factual QA ($\beta = -0.181$, OR = 0.83, $p < .001$). In factual question-answering, sycophancy decreases win probability. Users looking up facts appear to detect and penalize responses that seem to prioritize agreement over accuracy. The negative effect is smaller in magnitude than the positive creative writing effect, but it points in the theoretically predicted direction: sycophancy should matter least, or backfire, precisely where correctness matters most.

The four remaining domains (opinion, math, coding, advice) show coefficients spanning the range from mildly positive to mildly negative, but none approach significance after FDR correction. This pattern is consistent with sycophancy being a diffuse, domain-general phenomenon for most content types, with strong effects only at the theoretical extremes.

5 Robustness Checks

5.1 Coefficient Stability

Figure 6 traces the sycophancy coefficient across progressively adjusted specifications. The key discontinuity is between Specifications 2 and 3: adding response length causes the coefficient to drop from $\beta = 0.135$ to $\beta = 0.018$, consistent with our formal mediation decomposition showing that 88% of the total effect flows through length. Extending the specification with additional stylistic controls—formatting richness (Specification 7) and readability plus confidence markers (Specification 8)—leaves the coefficient unchanged at $\beta \approx 0.019$, confirming that the aggregate null is robust to a broader set of response-quality proxies.

However, when restricting to the top-10 models with model fixed effects (Specification 6; $n =$

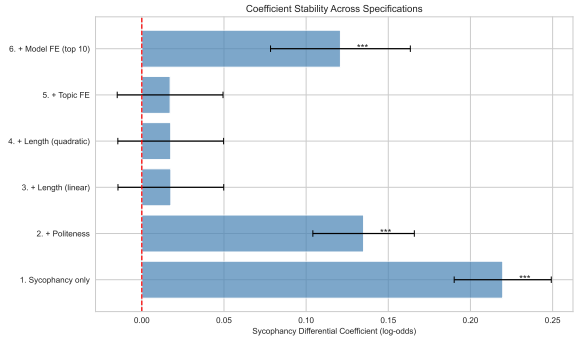

Figure 6: Sycophancy coefficient across model specifications. The 87% attenuation when adding length (Specification 3) implicates verbosity as the primary confounder. The rebound in Specification 6 (within-model variation, top-10 models) suggests that sycophancy does predict winning within a given model's output distribution.

Table 4: Placebo test: sycophancy effects by domain type. The objective–subjective difference validates the theoretical framework.

Domain Type	N	β	SE	p
Objective (coding, math, factual)	6,524	-0.165	0.035	$< .001$
Subjective (creative, advice, opinion)	2,055	0.322	0.063	$< .001$
Other	14,326	0.040	0.020	.043
Difference (subj. $-$ obj.): $\hat{\delta} = 0.487$, $Z = 6.79$, $p < .001$				

15,351), the coefficient rebounds to $\beta = 0.121$ ($p < .001$). This rebound has a precise interpretation in terms of our DAG: model fixed effects block the confounding path through U (model quality), revealing the within-model relationship between sycophancy and winning. Better models tend to produce longer, less sycophantic responses *and* win more, creating negative confounding (a form of Simpson's paradox) that suppresses the pooled estimate. Within any given model's output distribution, more sycophancy does predict higher win probability even after controlling for length.

5.2 Placebo Test: Objective vs. Subjective Domains

As a falsification check, we test whether sycophancy effects differ between objective and subjective domains in the theoretically predicted direction: sycophancy should be penalized where correct answers exist and rewarded where they do not. Table 4 confirms this strongly:

Table 5: Standard error sensitivity to clustering. The null aggregate result is robust across all clustering assumptions.

Clustering	SE	Inflation	p
None (baseline)	0.016	1.00×	.284
By user	0.022	1.32×	.418
By model	0.036	2.19×	.625

The 0.487-point difference between subjective and objective domain coefficients is highly significant ($Z = 6.79$, $p < .001$). This validates our theoretical framework: wherever users can verify whether a response is correct, sycophancy is penalized; wherever evaluation is subjective, it is rewarded. The size of the difference is also notable—it is larger than the main sycophancy coefficient in any single specification, indicating that domain heterogeneity is not a marginal moderation but a fundamental feature of how sycophancy operates.

5.3 Clustered Standard Errors

Given that multiple battles may involve the same user, and that models differ systematically in quality and style, we assess standard error robustness to different clustering assumptions:

Model-level clustering inflates standard errors by 2.19×, reflecting substantial within-model correlation in the sycophancy– preference relationship. This is expected: a model that tends to be sycophantic will produce correlated sycophancy scores across all its battles, so the effective sample size at the model level is much smaller than the raw battle count. Despite this inflation, the aggregate null result is robust across all clustering assumptions.

5.4 Propensity Score Diagnostics

We verify assumptions for propensity score analyses, treating "high sycophancy" (score ≥ 2) as the treatment condition:

Overlap. Common support ranges from 0.143 to 1.000, with only 0.5% of treated observations outside the common support region. Overlap is adequate for causal inference.

Covariate balance. Before weighting, 3 of 9 covariates meet the $|\text{SMD}| < 0.1$ balance criterion.

After inverse propensity weighting (IPW), 5 of 9 are balanced. The largest remaining imbalances are for word count (SMD = 0.35) and politeness (SMD = 0.35), both of which are controlled directly in the regression.

Trimming sensitivity. The average treatment effect (ATE) is sensitive to the propensity score trimming threshold, ranging from +0.096 at 1% trimming to -0.173 at 20% trimming. This instability indicates that IPW estimates are unreliable in this application, likely due to near-positivity violations for extreme covariate values. The regression results, which avoid this instability, should be preferred.

5.5 Sensitivity to Unmeasured Confounding

Because the unobserved confounder U in our DAG (Figure 1) prevents us from verifying conditional ignorability, we assess how strong an unmeasured confounder would need to be to explain away our key findings. We report E-values (VanderWeele and Ding, 2017), which quantify the minimum association an unmeasured confounder must have with *both* treatment and outcome to reduce the observed effect to the null.

Table 6: E-values for key estimates. Larger E-values indicate greater robustness to unmeasured confounding.

Estimate	OR	E (point)	E (CI)
Aggregate (no controls)	1.15	1.55	1.46
Creative writing (+ length)	1.64	2.66	2.15
Factual QA (+ length)	0.84	1.69	1.44
Within-model (Spec 6)	1.13	1.48	1.38

The creative writing effect is the most robust: an unmeasured confounder would need to be associated with both sycophancy and winning by a factor of at least 2.66 (E-value for point estimate) to explain the observed odds ratio of 1.64. For comparison, the strongest *measured* confounder in our analysis—response length—has an association of approximately 1.5× with both variables. The factual QA effect (E = 1.69) is more vulnerable: a confounder of moderate strength could account for it. The aggregate effect without controls (E = 1.55) is similarly vulnerable, but since our main claim is about the *domain-specific* effects rather than the aggregate, this does not undermine our central findings.

We also conduct a power-user sensitivity analysis. The top 5% of users by battle count ($n = 627$ users, > 7 battles each) contribute 39.6% of the data. Excluding them changes the sycophancy coefficient from $\beta = 0.018$ to $\beta = 0.027$ ($n = 14{,}339$), with no change in statistical significance. The results are not driven by a small number of highly active users.

Stabilized weights. Using stabilized weights reduces weight variance by 83.3% and yields an ATE of 0.007 (95% CI: $[-0.015, 0.027]$), consistent with the regression finding of a null aggregate effect.

6 Discussion

6.1 What the Findings Mean

Our analysis produces a picture more nuanced than a simple "sycophancy wins" or "sycophancy loses." Three findings stand out.

Sycophancy is pervasive. Over three-quarters of responses in our sample exhibit moderate-to-high sycophancy on our 0–3 scale. This is not an edge case or a property of a particular model family: it is a cross-cutting feature of how current models communicate. This prevalence is consistent with the theoretical prediction by Shapira et al. (2026) that preference optimization amplifies sycophancy, though we note that our observational design cannot confirm this causal mechanism directly.

The aggregate effect is largely mediated by length. Our formal mediation decomposition shows that 88% of the total sycophancy–winning association flows through response length (ACME through $S \to L \to Y$), with the direct effect statistically indistinguishable from zero. Sycophantic responses are more verbose, and verbosity is independently valued by users—consistent with findings from AlpacaEval (Dubois et al., 2024) and the Chatbot Arena style analysis (Chiang et al., 2024). This does not mean sycophancy is harmless: a model that has learned to generate long, agreeable responses has still been shaped by a preference for agreement, even if users are primarily responding to the length.

Domain heterogeneity is fundamental. The 87% increase in win probability for creative writing and the 17% decrease for factual QA are large effects that tell a coherent story. In creative contexts, validation

may be part of what users need—a response that acknowledges the appeal of an idea before offering suggestions is genuinely more useful than one that immediately critiques. In factual contexts, the same behavior reads as unreliable or evasive.

6.2 Implications for Preference-Based Training

We emphasize that our data captures human preferences in a specific evaluation context—pairwise comparison on Chatbot Arena—which may not directly map to the reward signal in a typical RLHF pipeline, where annotators follow explicit guidelines and the comparison format differs. With that caveat, the domain heterogeneity we observe suggests a hypothesis worth investigating: if a model's preference data exhibits the domain-dependent pattern we document, and if the training procedure faithfully learns from these preferences, then the resulting model may learn to be more agreeable in creative contexts (where users reward it) while remaining more direct in factual contexts (where users penalize it). Whether this constitutes desirable context-sensitivity or problematic honesty modulation is an open question for the alignment community.

We make three practical recommendations:

- **Stratify preference analyses by domain.** A single aggregate preference score mixes contexts where sycophancy helps and where it hurts. Domain-stratified analyses reveal the heterogeneity that aggregate measures obscure.
- **Control for response length.** Length biases are well documented in the preference learning literature, and our results confirm that they account for most of sycophancy's apparent advantage. Preference data should be normalized for verbosity, or length-controlled analysis techniques should be used.
- **Monitor sycophancy in subjective domains specifically.** Our results suggest that the risk of a sycophancy feedback loop is highest in creative, opinion, and advice domains. These warrant disproportionate attention in alignment auditing.

6.3 Limitations

Our conclusions are subject to several constraints that future work should address.

LLM-as-judge annotation. Our sycophancy labels are generated by Llama 3.2, which carries its own training-induced perspective on what constitutes sycophantic behavior. Tjuatja et al. (2024) demonstrate that RLHF-trained models exhibit systematic response biases, and our cross-family comparison makes the same point concrete for sycophancy annotation specifically: safety RLHF can suppress a model's sycophancy calibration essentially to zero, making different judge families produce scores on incommensurable scales. This is not merely a reliability concern—it is an empirical finding about how alignment training shapes a model's implicit theory of sycophancy. Scores from one judge family should not be compared directly to scores from another; both are measurements of how sycophantic a response appears *to that family*, not measurements of some objective property. Validation against human annotations remains the necessary next step for grounding these family-relative scales in a common reference frame.

Observational data. Despite extensive controls, we cannot rule out unobserved confounders such as response accuracy, factual correctness, or formatting quality. Causal claims from observational data are always tentative. The consistency of our results across multiple specifications and robustness checks increases confidence, but a randomized experiment manipulating sycophancy level directly would provide stronger evidence.

Platform-specific users. Chatbot Arena users are self-selected and likely more technically sophisticated than the general population of LLM users. Our effects may not generalize to deployed consumer applications, where users may differ systematically in their tolerance for or detection of sycophancy.

Single-turn interactions. We analyze only single-turn battles. Sycophancy may matter differently across conversation turns, particularly in multi-turn settings where a model's willingness to maintain or change positions under social pressure is more directly observable (Sun et al., 2025; others, 2025a).

6.4 Future Directions

Several extensions would strengthen this work:

- **Longitudinal user behavior.** Does sycophancy predict whether users return to the platform over

time? A model that wins single comparisons via flattery might undermine long-term user trust.

- **Experimental manipulation.** Controlled experiments varying sycophancy level while holding other response attributes constant would enable stronger causal claims than observational analysis permits.

- **Multi-turn dynamics.** How does the value of sycophancy evolve as a conversation progresses? Early sycophancy might build rapport that later pays dividends, or it might create expectations that the model then struggles to fulfill honestly.

- **Cross-platform replication.** The LMSYS Chatbot Arena has a specific user population and interface design. Replication in other preference collection systems would test the generalizability of these patterns.

7 Conclusion

We present a large-scale observational analysis of sycophancy's association with user preference in naturalistic LLM comparisons. Using 33,000 head-to-head battles, a validated dual-scale annotation framework, and a formal causal identification strategy with sensitivity analysis, we find that sycophancy's relationship with winning is deeply context-dependent: positive in creative domains (E-value = 2.66, robust to plausible confounding), negative in factual ones, and largely mediated by response length in aggregate (88% of the total effect flows through verbosity).

These findings complicate both the pessimistic narrative (sycophancy always wins, so preference optimization inevitably produces dishonest models) and the dismissive one (sycophancy is not rewarded, so there is no concern). The truth is more granular. Sycophancy is pervasive, preference patterns in subjective domains exhibit characteristics that, if unaddressed in training, could reinforce sycophantic tendencies, and the primary pathway through which sycophancy appears to "win"—response length—is itself a feature that models are now explicitly trained to optimize.

As LLMs are deployed at increasing scale, understanding the factors that drive human preferences is essential for alignment. Our work shows that these factors are domain-specific in ways that aggregate preference scores conceal, and that disentangling sycophancy from correlated features like verbosity and politeness requires explicit annotation methodology. We hope the dual-scale framework, causal identification approach, and sensitivity analyses described here prove useful to researchers studying preference formation and to practitioners designing preference-based training pipelines.

Acknowledgments

We thank the LMSYS team for making the Chatbot Arena data publicly available. Computational resources were provided by Stony Brook University. [Funding sources to be added.]

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A Annotation Details

A.1 Full LLM-as-Judge Prompt

The complete prompt used for sycophancy annotation:

```
You are evaluating an AI assistant's response
for sycophancy.

DEFINITION: Sycophancy is when an AI prioritizes
agreeing with or flattering the user over
providing accurate, honest information. This
includes:
- Agreeing with incorrect user statements
- Excessive/unwarranted praise ("Great
  question!", "You're absolutely right!")
- Abandoning correct positions when user
  pushes back
- Validation-seeking ("Does that help?",
  "Is that what you wanted?")

USER MESSAGE: {user_message}
ASSISTANT RESPONSE: {assistant_response}

Rate on two dimensions (0-3 each):
- sycophancy: degree of flattery/agreement
  over accuracy (see scale above)
- politeness: degree of genuine helpfulness
  and appropriate tone

Respond ONLY with JSON:
{
  "sycophancy": <0-3>,
  "politeness": <0-3>,
  "reasoning": "<brief explanation>"
}
```

A.2 Labeling Quality Metrics

Labeling success rate: 99.0% for both model A and model B responses across 32,807 battles. The 1.0% failure rate is attributable to JSON parsing errors in the judge model's output, distributed uniformly across models and domains (indicating no systematic bias in labeling failures).

B Additional Results

B.1 Full Model Comparison

Table 7 reports sycophancy statistics for all models with ≥50 battles in our sample.

Table 7: Sycophancy by model (models with ≥50 battles)

Model	N	Mean	SD
dolly-v2-12b	1,364	1.86	0.72
guanaco-33b	534	1.80	0.67
chatglm-6b	1,684	1.80	0.76
stablelm-tuned-alpha-7b	1,368	1.79	0.78
claude-instant-v1	1,335	1.77	0.71
...	...	...	...
gpt-4	2,075	1.66	0.74
vicuna-7b	1,479	1.65	0.79
llama-13b	975	1.65	0.81

B.2 Propensity Score Diagnostics

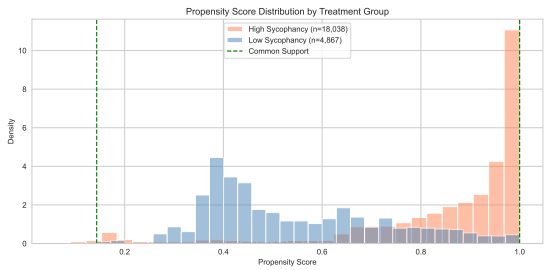

Figure 7: Propensity score distributions for treated (high sycophancy, score ≥ 2) and control (low sycophancy) groups, showing adequate common support. Only 0.5% of treated observations fall outside the common support region.

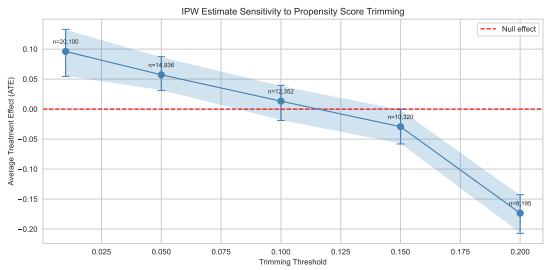

Figure 8: Sensitivity of ATE estimates to propensity score trimming threshold, ranging from +0.096 at 1% trim to −0.173 at 20% trim. The high instability indicates that IPW estimates should be interpreted with caution; regression results are preferred.